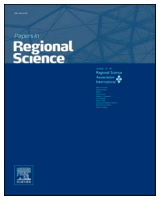
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Regional institutions and the urban digital divide

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ABSTRACT

The digital divide represents one of the many facets of socio-economic inequalities, creating a wedge between citizens with the means and ability to reap the benefits related to modern information and communication technologies (ICTs) and those who lack these resources and skills. Among its determinants at the individual and more aggregate level, space-specific factors, including the quality of regional institutions, may affect its geographical breakdown. In this paper we exploit data covering 181 European cities for the period 2008–2017, examining the spatial distribution of the within-city digital divide and how this relates to several local characteristics. In particular, we focus on the interplay between geographic variables and institutional quality at the local level, measured with the University of Gothenburg's Quality of Governance dataset. Within-city digital divide is computed using individual-level data on European citizens, thus controlling for individual traits that the literature has identified as relevant determinants. These include gender, age, education level, and occupational information, among others. Our paper sheds light both on the individual level determinants of the within-city digital divide, and considers how the uneven spatial distribution of the urban digital divide is related to local institutional, economic and social characteristics. Implications of our work can help inform local policy makers, who increasingly rely on digital solutions in urban policy planning, to consider the existence of a local digital divide among citizens and adopt corrective measures to help achieve digital urban policies' full potential. Our empirical results document a negative association between the quality of local institutions and the within-city digital divide. Results are robust to a number of alternative specifications, as well as to the use of historical instruments.

1. Introduction

Digital technologies are disrupting traditional manufacturing techniques (Llopis-Albert et al., 2021; Dyba and Di Maria, 2024), and changing the way products are designed, assembled, sold, and distributed with an intensity that made many talk about a new technological revolution, some even calling for an original paradigm shift, defined as *Industry 4.0* (Lasi et al., 2014). At the same time, these technologies are also impacting how consumers and citizens live, work, learn and consume. Within the universal role played by digital technologies as a labor-replacing and overall life-disrupting factor, Europe presents substantial spatial variations in the digital skills of the labor force and the general population (Vicente and López, 2011; Gomes and Dias, 2025).

According to recent EUROSTAT regional statistics, in 2022, while on average in the European Union only 7 per cent of citizens from 16 to 74 years of age never used the internet, in some regions located in Greece, Italy, Portugal, Bulgaria, and France this same statistic ranged between 14 and 20 per cent (EUROSTAT, 2024). Along the same lines, in some regions such as Luxembourg and Ireland's Eastern and Midland, roughly half of the population aged 16–74 has ordered goods or services over the internet from sellers in other EU Countries; while the same percentage is on average equal to 17 per cent for the EU as a whole. These statistics document significant spatial variations in the diffusion of digital technologies uptake even within the EU27, which is – compared to other institutional settings – relatively less diverse in terms of levels of economic development.

The uneven diffusion of digital technologies (in terms of their availability, their use, and their potential impact) has been termed *digital divide*. While the exact first date of appearance of this definition is not universally shared, according to Bulger (2007), this is first mentioned in

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a 1996 Los Angeles Times article by Amy Harmon discussing the early diffusion of the internet in the first half of the 1990s. Once the internet extended outside the US and became a global phenomenon, with a further leap caused by the introduction of laptops and smartphones, allowing the connection to the network from remote, the scientific debate focused on the reasons for some people, firms, and areas to be relatively less prone to adopt digital technologies, thus causing the rising tide of digital divide. Early studies provided cross-Country empirical evidence about the strength of the digital divide, and its main determinants.[3] All in all, cross-country evidence suggests a positive effect of institutional quality in reducing the digital divide, where the quality of institutions is thought of as how well formal institutions create and promote an environment that represents a level-playing field for all economic actors. Empirically, institutional quality in this sense can be measured by considering if the public administration is efficient, impartial, non-corrupt and is perceived as such by its citizens.[4]

The microfoundations of the institutions-digital technologies link are complex. On the supply side, a major role in enhancing the diffusion of digital technologies has traditionally been played by public procurement. Vast evidence in this field is summarized in (Sanchez-Graells, 2024), which discusses the role of national and local government in fostering the digital transition, with a particular acceleration during the pandemic, and based on the diffusion of AI. Higher quality institutions are also expected to deliver more effectively and less expensively the infrastructure needed for digital technologies, although the latter can at times help filling institutional voids and acting as market-makers.

On the demand side, institutions fall within the perimeter of market incentives for firms to adopt digital technologies in the empirical assessment by (Nicoletti et al., 2020). In their work, they find that better institutions can help reducing "*[…] administrative burdens on start-ups, barriers to competition in services and [enhance] access to risk capital*" (Nicoletti et al., 2020, p. 7). An evolutionary perspective in stressing the role of institutions in enhancing (or hampering) ICT diffusion is at the basis of the empirical analyses presented in Cirillo et al. (2023). Their results provide a nuanced picture of the modes of adoption of digital technologies in a large sample of Italian firms, showing heterogeneity along the size, industry, human capital, skill-content, and institutional axes, much like in the case of other space-specific factors (Rodríguez-Pose et al., 2024).

Looking at the academic literature on the digital divide and its determinants, as the diffusion of digital technologies became pervasive, the interest in the problem of digital divide abated. At this stage, little attention was paid to two factors that may contribute to our understanding of the digital divide. On the one hand, intra-national disparities have been discussed comparably less (Vicente and López, 2011), with a few notable exceptions (Szeles, 2018; Caragliu and Del Bo, 2023; Lucendo-Monedero et al., 2019 and Reggi and Gil-Garcia, 2021). On the other hand, insufficient attention has been paid to causal mechanisms, as regions and cities that are characterized by better institutional settings could be structurally more prone to keep the digital divide under control, for instance due to a common factor driving both outcomes.

In this paper we address this gap, and in particular provide three main advances w.r.t. the literature:

- We present a novel measure of the interpersonal digital divide within cities based on individual level data on possession, use and impact of digital devices;
- We bridge the cross-country literature on the role of institutions in driving the digital divide with the recent focus on regional/urban level digital divide. In fact, to the best of our knowledge, this is the first work suggesting the existence of a role for regional institutions in driving the *urban* digital divide;
- We focus specifically on causality links, and exploit exogeneous variations in historical instruments to separate out long-run determinants of regional institutions from their effects on the urban digital divide, thus allowing for safer inference on the causal impact that better institutions may have on the reduction of the digital divide.

This translates into the following research question:

RQ	*Does the quality of regional institutions reduce the urban digital divide?*

To answer this question we proceed as follows. Section 2 briefly discusses the related literature, with a specific focus on the role of formal institutions in driving the digital divide, specifically at the local level. Section 3 presents our empirical specification and the data collect to answer our research question. Section 4 provides a detailed account of the historical instruments that define our identification strategy. Section 5 presents our empirical findings. Finally, Section 6 concludes and draws some policy implications stemming from our results.

2. Do local institutions influence digital outcomes?

Despite the relevance of digital skills in driving several economic outcomes (Hulten, 2017), the empirical literature analyzing the determinants of the digital divide is relatively scant, and the evidence is even thinner when looking at studies that deal with its regional breakdown.

In fact, in a first wave of studies, the scientific debate revolved around cross-country analyses of the aggregate level of digital divide. In this branch of the literature, Chinn and Fairlie (2007) find that, at the Country level, and focusing on the penetration rate of digital technologies, the main factors determining the spatial distribution of digital technologies included economic, demographic, and infrastructure variables, as well as telecommunications pricing measures, and regulatory quality.

Along the road, micro and local data on the diffusion of ICTs at the local level became available, so that the intensity of the digital divide at the regional and urban level could be measured and mapped. For instance, Lucendo-Monedero et al. (2019) documents that the digital divide among top and worst performers among European regions is around 37 per cent, with spatial autocorrelation analysis showing a North-West to South-East trend in the divide. Similar regional disparities in the diffusion and uptake of digital technologies have been documented for Canada (Koch, 2022), China (Wang et al., 2021), and the US (Chakraborty and Bosman, 2005).

As gradually ICT availability and use extended to developing Countries, spatial variations in infrastructure gradually eroded, mostly thanks to the vast diffusion of mobile networks to developing countries (Rouvinen, 2006). However, the digital divide did not disappear, suggesting the need to explore other potential determinants other than physical availability. One strand of literature focused, at the national level, on the role of institutional quality. An early example of a study documenting the role of a specific facet of institutional quality, namely regulation, is by Chinn and Fairlie (2007), while other evidence on the effect of institutional stability and quality on decreasing the digital divide is presented, among many, in Skaletsky et al., (2016), Pick et al. (2021), Skaletsky et al. (2016), and Pick et al. (2021).

From an a-spatial angle, different arguments have been proposed to

[3] In this line, substantial information on the extent of the digital divide within and between European Union (henceforth, EU) Countries has been provided by the EU's Digital Economy and Society Index (DESI). This run from 2014 to 2022, while since 2023 DESI has been incorporated in the State of the Digital Decade report.

[4] Recent evidence has also been emerging on the role of institutional quality in driving the effectiveness of territorial reforms reshaping administrative boundaries. See e.g. Ritter (2025).

explain the role of institutions in driving the intensity of the regional divide. Several studies have documented how different dimensions of institutional quality and change are relevant determinants of the digital divide. Milner (2006) argues that along with technological and economic factors, political institutions may be the single most relevant factor explain spatial differentials in digital technology uptake. At the Country level, stakeholders who believe they would lose from the diffusion of ICTs actively attempted to slow down the diffusion of digital technologies. These include autocracies, or governments with limited accountability and poor popular support. Billon et al. (2009) find that government effectiveness is a significant determinant of the digital divide in a cross-country multivariate analysis for global countries. More recently, Rubino et al. (2020), focusing on European countries, show that there is a significant relationship between country-level cultural institutional setting, as in the Hofstede framework, and the national digital divide. Considering instead political institutional quality, namely democratization, Coccia (2020) documents that institutional change, in the direction of increasing democratization of countries, is a one of the factors behind the creation and diffusion of innovations, thus decreasing the existing digital divide.[5]

Evidence on this point is actually mixed. Some Countries classified as authoritarian regimes according to *The Economist*'s Democracy Index (Economist Intelligence Unit, 2023) are characterized by large diffusion of digital technologies (see e.g. China), while others keep applying restrictive policies on the use of some ICTs and digital platforms such as social media. For instance, following its aggression on Ukraine, Russia banned some social media (such as Facebook, Twitter, and Instagram) allegedly responsible for anti-regime propaganda (San Francisco, 2022). Empirical evidence suggests that indeed over the time span we observe with our data some degree of catch-up of autocracies in the use of digital technologies seems to have been taking place (Stier, 2017; Putnam, 2000). An implicit corollary of this argument is that strong, democratic institutions would lead to a better regulatory framework, capable of enhancing the diffusion of ICTs, and the way these are exploited by consumers and firms (Srinuan and Bohlin, 2011).

Recent empirical evidence studying the main determinants of the *regional* digital divide in Europe include Szeles (2018) who finds that policies fostering regional economic growth, increasing the tertiary education attainments and R&D, and providing incentives against school dropouts are the most effective tools to reduce regional digital divides. More recently, Caragliu and Del Bo (2023) focused on the role of urban smartness in reducing the digital divide, further distinguishing between ownership from use effects (i.e., first-order and second-order digital divide). This distinction also lies at the roots of the empirical findings discussed in Castellacci et al. (2020), who suggest that the local endowment of digital skills facilitates branching of new technologies. At the urban level, most studies have focused on the urban-rural digital divide, while not considering explicitly other determinants which might affect this aspect among cities with similar urbanization characteristics. Recent examples of this strand include de Clercq et al., (2023); Thomä, (2023) and (Peng and Dan (2023). Within this subnational framework, relatively recently a debate on the role of local institutions in driving regional outcomes, especially for the European case, emerged, based also on the availability of data on the quality of regional institutions collected by the University of Gothenburg (Charron et al., 2014; Charron et al., 2015).[6]

To sum up this brief review, at the current stage of the scientific debate, two issues appear relatively underexplored, thus calling for further research. On the one hand, the substantial spatial variation in regional institutions has not been explicitly a focus of prior empirical work, although this literature is growing also thanks to the availability of regional level data on institutional quality (Charron et al., 2015).[7] This dataset explicitly measures, at the regional level in Europe, the perception of citizens on the quality, impartiality and corruption of public services, providing for the first time a subnational picture of the variation of institutional quality in European regions. Information is collected by administering a survey to more than 129,000 EU citizens whose location is encoded for 208 NUTS1 and NUTS 2 regions. Questions focus on the citizens' perception of the quality of their local institutions. In particular, questions revolve around "*impartiality (i.e. that the government upholds an impartial treatment of all citizens irrespective of their personal characteristics or connections), corruption (i.e. that there is no abuse of public office for private gain), and quality (i.e. that the public services as perceived as high-quality)*" (Charron et al., 2021, p. 3).

Much prior empirical literature has at least partially neglected a focus on the causal mechanisms that drive the association between institutions and the digital divide. Within this framework, the European case study appears particularly promising in filling both gaps, as it presents, as above anticipated, substantial variations in both regional institutions and the regional and urban digital divide.

From an operational perspective, we fill the first gap by exploiting regional variation in institutions based on the University of Gothenburg's Quality of Governance dataset,[8] as well as in the intensity of the within-city digital divide. To deal with the second gap in the literature, we will resort to historical IVs.[9]

3. Empirical specification and data set

In order to explore whether institutional quality at the regional level reduces the urban level digital divide, we follow a two-stage procedure. In the first stage, we compute a measure of the digital divide at the urban level (within-city digital divide) with micro-level data focusing on the individual characteristics that shape the digital divide. In the second we aggregate this information at the urban level and explore its determinants, with a specific focus on the role of institutional quality, along with other urban and regional level determinants suggested by extant literature on the subject.

Our baseline reduced form specification regresses the intensity of city i's digital divide (DDi) on the quality of regional institutions ($INST_i$) and a number of regional and urban controls (Z_i) in a linear fashion (Eq. 1):

$$DD_i = \alpha + \beta_1 INST_i + \beta_2 Z_i + \varepsilon_i \quad (1)$$

The sample of cities analyzed potentially includes a grand total of

[5] Recent evidence would point at a prevailing role played by national institutions in driving regional economic development w.r.t. regional ones (Aparicio-Pérez et al., 2025). While this goes beyond the scope of our work, we believe this to be partially addressed by the inclusion of Country Fixed Effects in our regressions. At the same time, this may well represent a promising avenue for future research.

[6] Recent work in this one of research found evidence on the role of regional institutions in mediating the impact of digital technologies on regional income inequalities (Antonietti et al., 2025)

[7] Maps, and the main descriptive statistics on the extent of variation in the urban digital divide and the quality of regional institutions, are presented in Technical Appendix A.1. Both maps and the descriptives suggest that a very relevant role in the explanation of overall EU-wide variation is explained by the within Country component.

[8] This has become the standard tool for measuring regional institutions in the EU, and spatial variations in institutions have been employed in (Ganau and Rodríguez-Pose, 2019; Ketterer and Rodríguez-Pose, 2018), and (Rios, Beltrán-Esteve, Gianmoena, Peiró-Palomino, & Picazo-Tadeo, 2023) to explain various economic outcomes, including labor productivity, GDP growth, and women's empowerment.

[9] Details on the rationale for the use of historical IVs are presented in Section 4.

181 urban areas, whose complete list is shown in Technical Appendix A.2. List of urban areas in the data set.

To measure the intensity of the Digital Divide, we employ the measure first developed and proposed in Caragliu and Del Bo (2023). This is obtained as follows. First, an ancillary regression is built with the aim to predict the probability that individual users will adopt specific types of digital technologies. In this first stage regression (see Eq. (2) below), the intensity of digital divide is calculated as the within city standard deviation in the distribution of individual responses to the 2017 edition of the European Values Study (henceforth, EVS).[10] Individual responses are next predicted after regressing ICT ownership and usage against relevant individual traits. These include education, gender, income, age, student/employed/retired status, and Country fixed effects (Eq. 2):

$$prob(digital\ owenrship-usage_i)=\alpha+\beta_1*education_i+\beta_2*gender_i+\beta_3*age_i+\beta_4*working_status_i+\beta_5*\sum_{c=1}^{n}Country_{i,n} \quad (2)$$

For all 22,628 individuals for which all observations about information on individual characteristics and digital technologies uptake are available, we calculate the predicted linear probability of actually owning or using one of the digital technologies included in the questionnaire (desktop and laptop computers, tablets, smartphones, home internet connection, and standard mobile phone). The standard deviation of the predicted linear probabilities obtained from Eq. (2) is obtained as follows (Eq. 3):

$$DD_i=SD_c\left(\widehat{digital}_j\right)=\sqrt{\frac{\sum_{i=1}^{n}\left[\widehat{digital}_j-\overline{digital}\right]^2}{(n-1)}} \quad (3)$$

where $j = 1,\ldots, n$ indicates the number of individuals in each city i, and notations with a hat on top refer to values predicted in the instrumental regression in Eq. (2).

Going back to Eq. (1), the quality of regional institutions is measured with the University of Gothenburg's Quality of governance regional data base (Charron et al., 2014; Charron et al., 2015). The European Quality of Government Index (EQI) measures average citizens' perceptions and experiences with corruption, quality and impartiality of three essential public services – health, education and policing - in their region of residence, and is used as an indicator of the underlying regional institutional quality.[11] For NUTS2 regions in Europe, data comprise three pillars (quality, impartiality, and corruption) as perceived by EU citizens. These data are merged to our data set, which, as above anticipated, is collected for NUTS3 regions, proxying for metropolitan areas. This means that NUTS3 regions in our data set are assigned the score in the quality of governance data base characterizing the broader NUTS2 region to which they belong. With the exception of a few areas whereby the NUTS2 region overlaps with the NUTS3 one (e.g., Brussels), in most cases this implies a potential misalignment of the quality of institutions between the two geographies. However, this may represent a issue only if we had a prior (in either direction) on whether core cities within each NUTS2 region significantly differ from the rest of the NUTS2 region in terms of institutional quality. While, to the best of our knowledge, no other data base covering all EU27 Countries with the same level of information is available to capture the quality of regional institutions, indirect evidence suggests that no significant differences across NUTS3 regions belonging to the same NUTS2 region should affect our data.[12,13]

The Z vector includes instead space-specific factors that often found in the literature to be potentially conducive to a more (or less) extensive uptake of digital technologies. These include morphological characteristics, human capital and innovation factors, characteristics associated with urbanization economies, and the level of economic development.

Throughout the paper, our baseline geography is EUROSTAT's NUTS3 level, roughly corresponding to US Counties and Japanese prefectures. Despite heterogeneity in the average size of these regions, their definition is relatively homogeneous across the EU27,[14] and remains all in all constant over time, with a few changes that can be handled by means of EUROSTAT-generated crosswalks. NUTS3 regions represent a good balance in the trade-off between the level of spatial detail on the one hand, and the availability of data on the other hand. Besides, despite their definition being mostly based on administrative characteristics, their boundaries effectively capture commuting flows, thus proxying for extended local labor markets (Capello et al., 2022; Capello et al., 2023).

Data,[15] indicators, sources and time periods observed for the control variables in Eq. (1) are described in detail in Table 1.

The choice of the types of urban and regional characteristics we control for is based on the following rationale.

Morphological characteristics are expected to enhance, or hamper, an easier diffusion of digital technologies (Sant'Ana et al. n.d.). Areas more remote due to orographic characteristics (Sgroi and Modica, 2024), island location (Cave, 2012), and border areas (Hanclova, 2014), as well as sparsely populated regions all achieve low critical mass in terms of market size, thus making it more expensive to extensively cover their surface with ICT infrastructure.

Human capital and innovation are also critical factors in driving a society's uptake of ICTs. The complexity in the use of these technologies defines them as labor-augmenting technologies, but this requires skills in their use to fully discharge their positive impacts. In fact, increasing education and strengthening innovation systems are the two crucial factors in pulling lagging countries and regions closer to the technological frontier (Das and Drine, 2020).

The settlement structure of the areas we review for our analyses is also a relevant factor in driving digital technologies diffusion. In this sense, the nature of cities as a large market, and, more broadly, all the externality fields that make consumers and firms more productive due to their colocation (what in urban economics is termed agglomeration economies), make it more effective to concentrate public capital investment (Camagni, 1993), thereby making all else being equal less expensive to build digital infrastructure.

Lastly, a city's level of economic development is also a crucial prerequisite for digital technologies to be implemented (Kim, 2006).

As the outcome explained with Eq. (1) is calculated for 2017, Table 1 shows that in our analyses the data have been assembled with the goal to maximize the time lag between right- and left-hand side (thus minimizing the issue of possible reverse causation), while at the same time

[10] EVS' most recent version is in fact the 2017. In that edition, cooperation for joint data collection with the WVS started, but no additional information on pillars of the digital divide is available in later waves.

[11] https://ec.europa.eu/regional_policy/information-sources/maps/quality-of-government_en

[12] This evidence is presented in Technical Appendix A.1. Within-NUTS2 variation in regional quality of institutions between NUTS3 regions.

[13] Regional quality of institutions data have been frequently used at sub-NUTS2 level in several prior analyses. See e.g. (Iammarino et al., 2019; Audretsch et al., 2024 and Węziak-Białowolska, 2016), among many.

[14] According to Regulation (EC) No 1059/2003 of the European Parliament and of the Council of 26 May 2003, NUTS3 regions host populations ranging between 150,000 and 800,000 inhabitants. Their boundaries are continuously updated. Our empirical analyses are based on the 2016 version of NUTS3 regions.

[15] All collected for the NUTS3 level, with the exception of human capital and innovation measures, measured at the NUTS2 level, and assigned to each NUTS3 level region belonging to the larger NUTS2 spatial unit. At most, this choice should cause us to underestimate the intensity of R&D expenditure and tertiary education in our sample, as typically core urban areas tend to be characterized by values larger than their peri-urban and rural surroundings.

Table 1
Data, indicators, sources and time periods observed for the control variables in Eq. (1).

Type of control	Variable	Source of raw data	Time observed	Spatial breakdown
Morphological characteristics	Mountain regions	EUROSTAT regional typologies	2015 (Dijkstra and Poelman, 2015)	NUTS3
	Island regions			
	Sparsely populated regions			
	Border regions			
Human capital and innovation	Perc. tertiary education in aged 25–64	EUROSTAT	2000	NUTS2
	R&D intensity	EUROSTAT	Average 1989–2012	
Agglomeration economies	Dummy, = 1 if region hosts a Country capital	Own calculation	-	NUTS3
	Population density	EUROSTAT	Average 1989–2012	NUTS3
	Share of artificial land	Corine Land Cover	2009	NUTS3
Economic development	Log of urban GDP	EUROSTAT	Average 1995–2010	NUTS3

Source: Authors' elaboration

calculating long-period averages for regional and urban characteristics characterized by short-run variability (such as R&D activity, or urban GDP[16]). As these measures address the issue of endogeneity only partially, the next section is devoted to the way we aim at identifying estimated parameters.

4. Identification strategy

Our analyses may potentially suffer from endogeneity; in other words, the correlation observed through OLS estimates between regional institutions and the urban digital divide may be driven by other factors influencing both. In this case, the use of time lags on the right-hand side may not suffice to rule out the possibility. In our analyses, we opt for instrumental variables. In particular, we exploit exogeneous variation in two characteristics:

1. Historical transportation infrastructure;
2. Soft institutions.

The rationale behind the use of long lags of transportation infrastructure looks at the channels through which lower transport costs influence the formation of institutional quality (*validity* condition), without plausibly affecting the intensity of the digital divide at the urban level (*exclusion* condition).

For this empirical application, we sought variables that would be providing valid instruments. In particular, if we define X as the potentially endogenous variable (in our case, regional institutions), Z as the potential instrument (in our case, long time lags of transportations networks, and regional trust), and u as an error term, our instruments must satisfy the following conditions:[17]

Relevance: $Corr(X, Z) \neq 0$

Exogeneity: $Corr(Z, u) = 0$

To exemplify this with the first instrument, we first run an ancillary OLS regression of the form (Eq. 4):

$$INST_i = \beta_0 + \beta_1 * ROMAN_ROADS_i + v_i \quad (4)$$

where v_i is an i.i.d. disturbance within the ancillary regression.

We then calculate $\widehat{INST_i}$.as the predicted value in Eq. (4). Lastly, we run the main regression (Eq. 1), but replace the real vector of institutions with the value predicted from Eq. (4.):

$$DD_i = \alpha + \beta_1 \widehat{INST_i} + \beta_2 Z_i + \varepsilon_i \quad (5)$$

i.e., replacing the original institutions vector with the one predicted with the ancillary regression.

For starters, lower transport costs may increase the quality of institutions by fostering trade. Evidence in this sense is surveyed in Anderson and Van Wincoop (2004). Trade is also found to be positively associated with the quality of institutions (De Groot et al., 2004), although its relative role with respect of institutions in driving economic growth has been questioned in Dollar and Kraay (2003). A direct correlation between trade and infrastructure on the one hand, and the quality of institutions on the other hand, is discussed in Hochman et al. (2013), who theoretically model the role of infrastructure in enhancing aggregate trade, which has both a positive effect (through lower transport costs for exporting firms) and a negative one, due to the way transport infrastructure is financed by taxing firm activity. Lastly, while the empirical literature converges on the existence of a positive link between transport infrastructure and economic outcomes (Ferrentino and Vota, 2023; Santagata, 2022), a link between the former and the extent to which urban dwellers are unequally endowed with digital technologies seems untenable.

As for soft institutions, a literature spanning two decades around Robert Putnam's (and coauthors') work (Putnam et al., 1993; Putnam, 2000) on social capital found associations between these and institutional quality. For instance, Stiglitz (2000) argues that as countries develop, informal institutions (trust and networks) give way to more formal institutions in the form of agreements, rules, ad regulation, concerning the protection of property rights, impartiality, and the rule of law. The link between formal and informal institutions is discussed both theoretically and empirically in Casson et al. (2010); the evidence presented suggests the existence of a strong and persistent positive association between trust and networks on the one hand, and institutional quality on the other hand.[18]

Based on these strands of literature, we exploit three instruments to address endogeneity in the regional institutions-digital divide link.

For transport infrastructure, we employ the GIS of two substantial lags of the present day network. The first is the full network of Roman roads, while the second refers to the network of European railways as of 1870.

[16] For economic development, the shorter time series is due to the lack of homogeneous EUROSTAT GDP statistics at regional level, mostly for pre-accession Countries. We believe a 15 years period still represents a sufficient solution to the problem of short-run variability.

[17] (Stock and Watson, 2020), Ch. 12.

[18] The link between informal (trust) and formal (quality of governance) institutions needs not to be interpreted causally. For this instrument to be valid, it should be correlated with the instrumented variable, without however being correlated with the error term in the reduced form we estimate (Baum-Snow and Ferreira, 2015). Our findings provide support for this claim. First stage regressions, shown in Technical Appendix A.3. IV tests consistently find that trust is correlated with the regional quality of institutions; the causal direction of this relationship does not matter here. In fact, it could be well argue that hard and soft institutions are mutually connected by a biunivocal relationship (Muringani, 2022). Alternatively, we may assume trust to compensate for the lack of formal institutions (Tillmar and Lindkvist, 2007), or, by the same token, formal institutions to affect interpersonal trust (Raya and Resosudarmo, 2024).

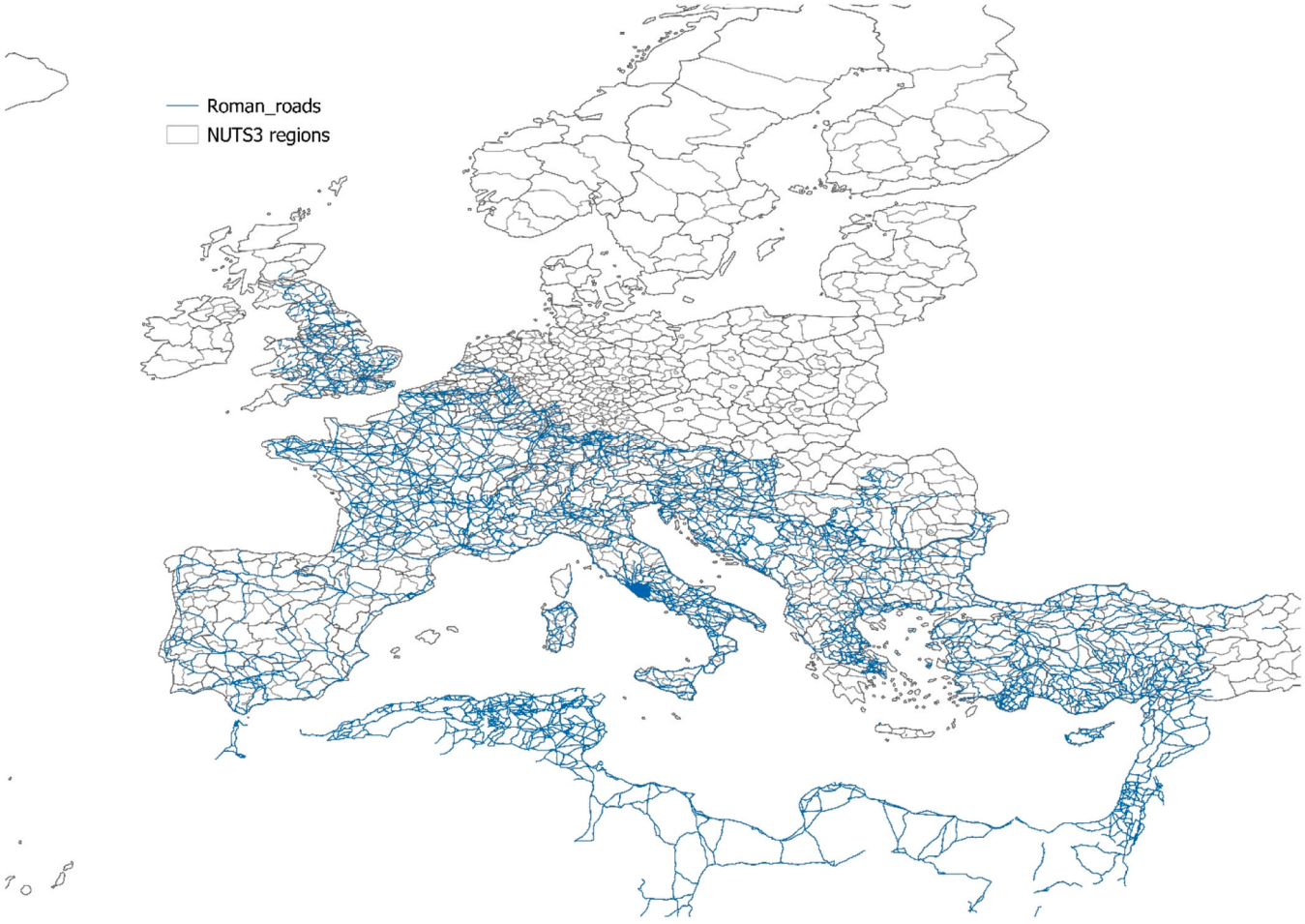


Fig. 1. Roman road network at the death of Trajan (117 CE). Source: Mc Cormick et al# (2013), Authors' elaboration.

The GIS of Roman roads has been first proposed in McCormick (2008). This data set, shown in the map in Fig. 1,[19] comprises the road network built during the last period of Rome's republic, to the era of the peak expansion of the empire (death of Emperor Trajan, 117 CE). The GIS comprises 7154 segments of Roman roads, covering 36 present-day Countries spanning Europe, Africa, and Asia.

Data have been recently employed in similar studies, such as De Benedictis et al. (2023), which suggests that the Roman transportation infrastructure crucially influenced the present-day network in Italy, thus making it a potentially useful way to address endogeneity in studies dealing with the impact of accessibility on various economic outcomes. Similar results about the persistency of transportation networks are also discussed in Dalgaard et al. (2022), who dig into the channels leading the Roman road network to shaping present-day urbanization patterns in the area formerly occupied by the Roman Empire. In fact, the use of the same roman roads network led Licio (2021) to propose a discrete version of the database as an instrument in growth regressions.

The second instrument using exogeneous variation in transport infrastructure is also based on a long time lag, related to 1870 railways. The GIS used in our work is first proposed in Morillas-Torné (2012), and first digitalized and employed in this same vein in Caragliu (2022). This second instrument is particularly helpful in that it covers areas in present day Central and Eastern European Countries (CEECs), which the Roman Empire never occupied, and, therefore, are not available in the previous network (Fig. 2).

Lastly, for soft institutions, we use the vector of regional trust generated by averaging out responses to the question "*Generally speaking, would you say that most people can be trusted or that you can't be too careful in dealing with people?*" within the 2017 edition of the *European Value Study* (henceforth. EVS). The EVS is the largest longitudinal data set collecting information on European citizens' perceptions and attitudes. Possible answers to the question include "Most people can be trusted" and "Can't be too careful". Our variable is the percentage of respondents "Most people can be trusted" for each NUTS3 region in our data base. The spatial distribution of available responses is shown in Fig. 3.

Fig. 3 shows a positive gradient for trust intensity in the direction of North-West. This statement is further evidenced by regressing this vector against the regions' centroids' coordinates (results shown for space limitations in Technical Appendix A.2), yielding a positive and significant association with the region's longitude, and a negative and significant one for the region's latitude.

5. Empirical results

Our results are discussed as follows. First, in Subsection 5.1 we show a baseline specification with only regional quality of governance being used as the explanatory factor for the urban digital divide. Subsection 5.2 shows a few robustness checks with the inclusion of additional RHS controls. Lastly, Subsection 5.3 verifies whether regional institution may have a nonlinear effect on the intensity of the urban digital divide.

[19] Across Figs. 1–3, the GIS of the instruments used is overlapped with EU NUTS3 regions' boundaries, representing our baseline geography.

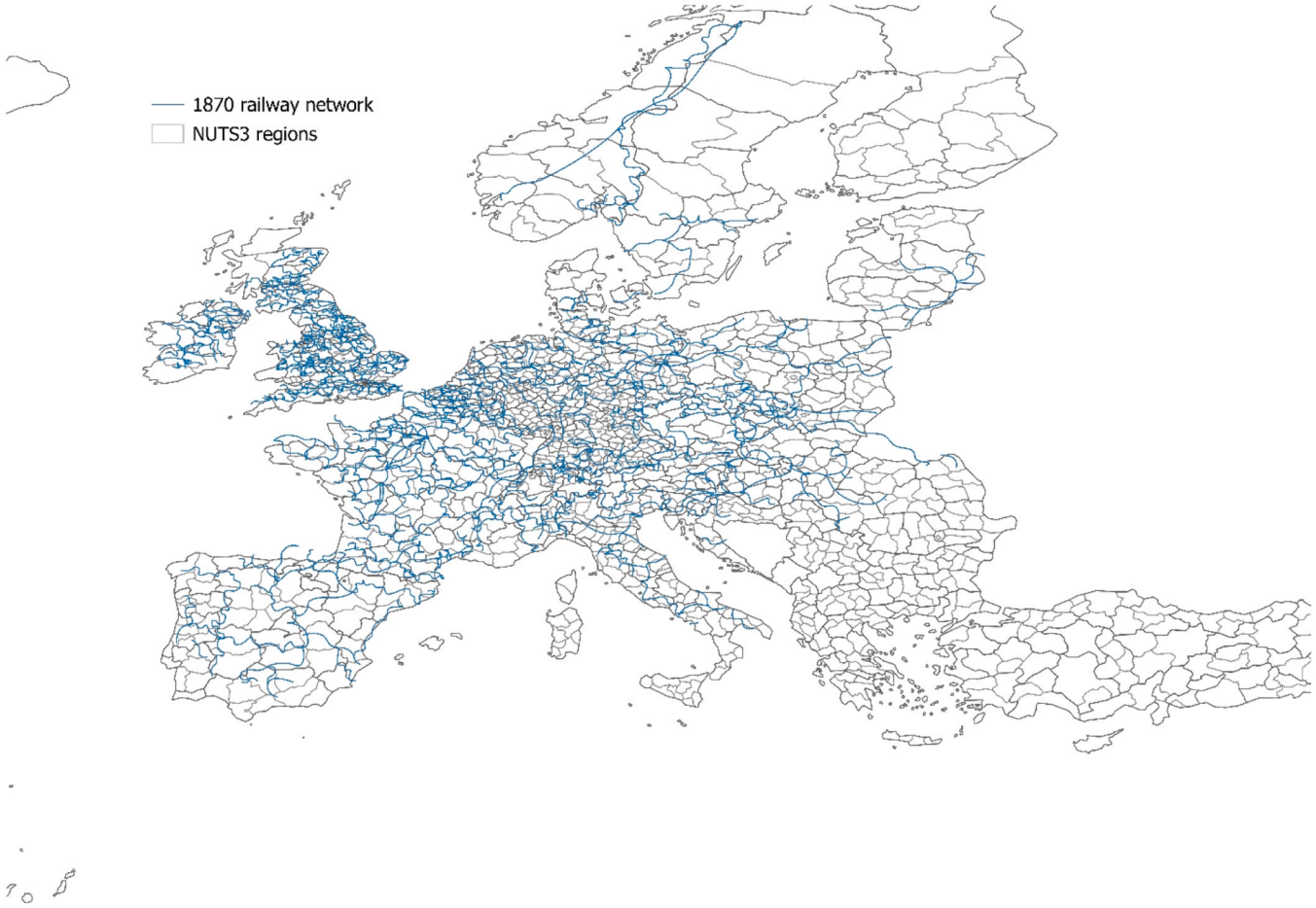


Fig. 2. European railway network in 1870. Source: Caragliu (2022), Authors' elaboration.

5.1. Baseline specification

This first subsection shows the results of regressing the urban digital divide on regional Quality of Governance only, without controls. Results are shown in Table 2 for standard heteroskedasticity-robust standard errors OLS estimates, while the same specification is estimated by means of Instrumental Variables in Table 3.

In both Table 2 and Table 3, the first column uses the aggregate indicator of regional institutional quality, while the second, third, and fourth column break this down into its three main pillars, viz. institutional quality, impartiality, and corruption. While these are arguably correlated with each other and in space, one may argue that some aspects of regional institutional quality may matter more for reducing the digital divide within cities.

Table 2 documents a rather stable negative and significant association between the quality of regional institutions and the urban digital divide. The estimated OLS parameter ranges between −.013 and −.018, with comparable significance levels. Pure quality of institutions and corruption stand out in terms of the percentage of linear variance explained (around.18 in both cases).

Table 3 replicates the same approach by means of instrumenting regional institutions with the three instruments described in Section 4, i.e. Trajan-period network of Roman roads; 1870 European railways; and regional trust. The drop in observations from OLS to IV regressions is due to the lower coverage of NUTS3 areas by our three instruments (jointly used).

Results in Table 3 provide robust evidence about the sign and statistical significance of the negative association between the quality of regional institutions and the urban digital divide.

Comparing the main regressor, i.e. the quality of regional institutions, across OLS (Table 2) and IV (Table 3) results, we find likely evidence of attenuation bias in the first set of results. This may be typically due to measurement error (quite likely, given the proxy nature of both our main RHS and the LHS variables), as well as to the presence of random errors (i.e. random fluctuations in either variable). Indirect evidence of this issue is provided by the IV-based estimated parameters, suggesting that the real impact of the quality of regional institutions on the reduction of the urban digital divide may be roughly twice as large as the one found with OLS estimates.

For reasons of space limitations, and in order to enhance the readability of this section, the full battery of standard tests for underidentification, weak instruments, and overidentification are shown in Technical Appendix A.5. IV tests. In a nutshell, all across this first battery of regressions, we strongly reject (p-value=.000) the null hypotheses of the tests for Underidentification and the use of weak instruments, and do not reject the null for the Hansen J statistic, thereby suggesting that our baseline IV estimates are based on valid instruments (in other words, instruments uncorrelated with the error term of the main specification, while simultaneously holding that the excluded instruments are correctly excluded from the estimated equation).

5.2. Robustness checks

As the thee pillars of regional institutions quality behave relatively similarly across all specifications in Tables 2 and 3, we discard this nuance and from now on only focus on the aggregate indicator of regional institutions. This subsection focuses also on obtaining a richer picture of the relation between regional institutions and the digital

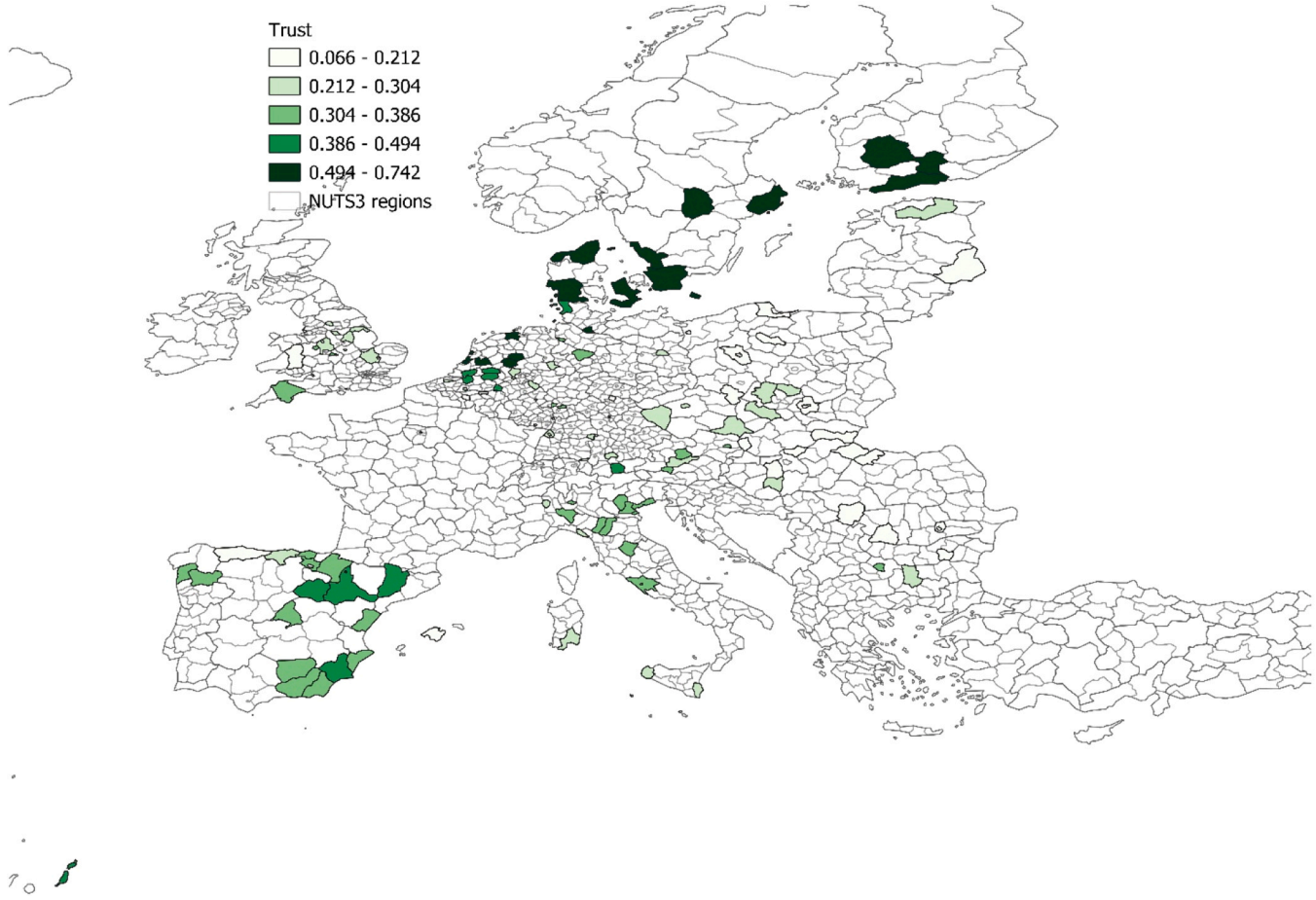


Fig. 3. Trust intensity in NUTS3 regions. Source of raw data: (EVS. 2020), Authors' elaboration.

Table 2
Baseline OLS estimates of the determinants of urban digital divide.

Model	*(1)*	*(2)*	*(3)*	*(4)*
Institutions specification	*Quality of Governance*	*Quality*	*Impartiality*	*Corruption*
Regional Quality of Governance	−0.0128***	-	-	-
	(−4.01)			
- Quality pillar		−0.0184***	-	-
		(−4.58)		
- Impartiality pillar	-	-	−0.0133***	-
			(−3.50)	
- Corruption pillar	-	-	-	−0.0162***
				(−4.27)
Constant term	0.211***	0.202***	0.202***	0.202***
	(94.59)	(46.46)	(42.40)	(46.22)
Observations	140	140	140	140
Estimation method	OLS	OLS	OLS	OLS
Adjusted R^2	0.061	0.138	0.081	0.135

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3
Baseline IV estimates of the determinants of urban digital divide.

Model	*(1)*	*(2)*	*(3)*	*(4)*
Institutions specification	*Quality of Governance*	*Quality*	*Impartiality*	*Corruption*
Regional Quality of Governance	−0.0367***	-	-	-
	(−5.26)			
- Quality pillar	-	−0.0353***	-	-
		(−4.33)		
- Impartiality pillar	-	-	−0.0361***	-
			(−4.34)	
- Corruption pillar	-	-	-	−0.0299***
				(−4.47)
Constant term	0.218***	0.201***	0.199***	0.201***
	(60.78)	(38.77)	(34.52)	(38.59)
Observations	107	107	107	107
Estimation method	IV			
Instruments	Trajan-period network of Roman roads; 1870 European railways; regional trust			
Adjusted R^2	−0.053	0.087	−0.082	0.071

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

divide, after controlling for other confounding factors that may potentially bias the relationship we measure. These controls follow the same structure as Tables 2 and 3: in Table 4 we show OLS results of regressing the digital divide on regional institutions, while also controlling for various regional typologies,[20] location characteristics, the city's level of human capital, its level of economic development as measured by urban GDP, and finally for possible Country heterogeneity. Table 5 replicates the same estimates on the basis of our IVs.

Table 4 provides robust evidence that our measure of regional institutions remains negatively and significantly associated with the intensity of the digital divide. The estimated parameter becomes increasingly larger in magnitude as we better identify it by means of including more controls. In Columns (4) and (5), the estimated

[20] "*Metropolitan regions are NUTS level 3 approximations of functional urban areas (composed of a city and its commuting zone) with at least 250 000 inhabitants*" (EUROSTAT, 2024).

Table 4
Robustness checks: OLS estimates of the determinants of urban digital divide.

Model	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*
Specification	*Regional typologies*	*Location characteristics*	*Human capital*	*Urban GDP*	*Country heterogeneity*
Regional Quality of Governance	−0.0135***	−0.0150***	−0.0208***	−0.0263***	−0.0250***
	(−3.91)	(−3.05)	(−3.56)	(−3.42)	(−3.23)
Metropolitan regions	0.000179	−0.000768	−0.00150	−0.00197	0.000191
	(0.07)	(−0.26)	(−0.51)	(−0.67)	(0.07)
Mountain regions	−0.00500***	−0.00508***	−0.00510***	−0.00472***	−0.00384***
	(−4.41)	(−4.11)	(−3.97)	(−3.59)	(−2.90)
Island regions	0.00203	0.000817	0.00146	0.0000457	0.00153
	(1.17)	(0.43)	(0.77)	(0.02)	(0.70)
Sparsely populated regions	0.0651***	0.0647***	0.0551***	0.0502***	0.0491***
	(5.71)	(5.43)	(4.43)	(3.77)	(3.60)
Border regions	−0.00270	−0.00234	−0.00190	−0.00178	−0.00137
	(−1.19)	(−1.10)	(−1.00)	(−0.92)	(−0.71)
R&D intensity	-	−0.0578	−0.249	−0.387	−0.292
		(−0.20)	(−0.88)	(−1.28)	(−1.09)
Dummy, = 1 if region hosts a Country capital	-	−0.00872	−0.0216**	−0.0226**	−0.0153*
		(−1.06)	(−2.44)	(−2.50)	(−1.70)
Population density		−0.00369	−0.00478	−0.00797	0.000937
		(−0.41)	(−0.52)	(−0.84)	(0.09)
Share of artificial land		0.000256	0.000263	0.000233	−0.0000370
		(1.20)	(1.12)	(0.99)	(−0.16)
Perc. tertiary education in aged 25–64			0.00159***	0.00161***	0.00111*
			(2.71)	(2.69)	(1.86)
Log of urban GDP				0.00999	0.00639
				(1.39)	(0.80)
Constant term	0.245***	0.247***	0.222***	0.128*	0.185**
	(32.65)	(27.28)	(19.31)	(1.81)	(2.34)
Observations	180	180	180	180	180
Country Fixed Effects	No	No	No	No	Yes
Adjusted R^2	0.158	0.155	0.182	0.183	0.271

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5
Robustness checks: IV estimates of the determinants of urban digital divide.

Model	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*
Specification	*Regional typologies*	*Location characteristics*	*Human capital*	*Urban GDP*	*Country heterogeneity*
Regional Quality of Governance	−0.0370***	−0.0434***	−0.0533***	−0.109***	−0.0949***
	(−5.12)	(−4.35)	(−4.80)	(−4.59)	(−4.45)
Metropolitan regions	0.0000171	−0.00148	−0.00210	−0.00325	0.000849
	(0.01)	(−0.40)	(−0.59)	(−0.78)	(0.22)
Mountain regions	−0.00431**	−0.00507**	−0.00424*	0.00165	0.00256
	(−2.33)	(−2.50)	(−1.91)	(0.43)	(0.68)
Island regions	0.00109	−0.00346	−0.00216	−0.0237**	−0.0201**
	(0.20)	(−0.60)	(−0.42)	(−2.38)	(−2.30)
Sparsely populated regions	0.0602***	0.0655***	0.0448***	−0.00888	−0.00293
	(5.13)	(5.20)	(3.32)	(−0.34)	(−0.11)
Border regions	−0.00139	0.000268	0.00139	0.00209	0.00223
	(−0.53)	(0.10)	(0.72)	(1.10)	(1.58)
R&D intensity	-	1.005**	0.562	−0.335	−0.103
		(2.11)	(1.22)	(−0.65)	(−0.24)
Dummy, = 1 if region hosts a Country capital	-	−0.0163	−0.0437***	−0.0722***	−0.0548***
		(−1.35)	(−3.45)	(−3.12)	(−2.58)
Population density	-	−0.0107	−0.0184**	−0.0540**	−0.0420**
		(−1.16)	(−1.97)	(−2.34)	(−2.12)
Share of artificial land	-	0.000396	0.000389	0.0000281	−0.000367
		(1.61)	(1.38)	(0.08)	(−1.07)
Perc. tertiary education in aged 25–64		-	0.00343***	0.00462***	0.00360***
			(3.70)	(3.64)	(3.32)
Log of urban GDP	-	-	-	0.0769***	0.0624***
				(3.15)	(2.85)
Constant term	0.249***	0.241***	0.181***	−0.404*	0.219***
	(18.87)	(15.49)	(7.70)	(−1.77)	(12.16)
Observations	131	127	127	127	127
Estimation method	IV	IV	IV	IV	IV
Instruments used	Trajan-period network of Roman roads; 1870 European railways; regional trust				
Country Fixed Effects	No	No	No	No	Yes
Adjusted R^2	0.065	0.055	0.074	0.004	0.287

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

parameter is found to be equal to around −2.5 percentage points.

In Table 5 the same structure is repeated for IV estimates. As documented in Subsection 5.1, the estimated parameters increase in size, while retaining the same order of magnitude and significance. In models (4) and (5), those with the preferred specification, a one per cent increase in the quality of regional institutions is found to be associated with a .1 per cent decrease in the urban digital divide.

In both Tables 4 and 5, three control variables call for specific comments, the first two being related to the morphological structure of the cities in our analyses. First, we find a positive and significant association between the sparsely populated regions dummy and the urban digital divide. This is expected, as a spatially uneven and low-density settlement structure arguably makes it more expensive to fill infrastructure gaps. The opposite result is found for mountain regions, despite these could also be considered as places where building digital infrastructure is expensive. This remains a possible puzzle to be addressed with future research. The third apparent puzzle in these results relates to the positive and significant association between the urban digital divide and the city's human capital (measured by the percentage of working age population with tertiary education). However, this link has already hinted at by prior evidence (see e.g. (Korupp and Szydlik, 2005)): the rationale behind this stylized fact is that workers with higher levels of education tend to complement more effectively capital, thereby widening the gap between low-skilled and high-skilled jobs, which, in turn, is reflected in this finding.

Detailed tests for underidentification, weak identification, and overidentification (shown for space limitations in Technical Appendix A.5. IV tests) provide overall evidence of the quality of the instrumentation strategy. Across all five models, underidentification and weak identification are supported by means of rejecting the standard null hypotheses of the Kleibergen-Paap rk LM statistic (underidentification test) and the Cragg-Donald Wald F-statistic (Weak identification test). Moreover, across models 1–3 and 5 no evidence is found for overidentification, while only for model 4 the Hansen J statistic is marginally rejected.

5.3. Nonlinear effects

Our last set of results is related to further robustness checks. Table 6 shows these results.

First, Columns (1) and (2) show the result of adding to our specification a dummy equal to 1 only for regions located in Central and Eastern European Countries (CEECs), first estimating the model with OLS, then with IVs. With the same structure, Columns (3) and (4) show the results of adding a control measuring the median age of the city population, first based on OLS, then on IVs. Estimated parameters remain in the interval −2 %–6 %.

Lastly, Model (5) adds a squared term for the regional institutions measure, to account for possible non-linearities which might go undetected by considering a linear model as in Eq. (1). As both the linear and the square terms are found to be negatively and significantly associated with the digital divide, our findings provide evidence on an increasingly larger impact of regional institutional quality on the urban digital divide. In this case we only show OLS results, as our instruments would

Table 6
Additional robustness checks: OLS and IV estimates of the determinants of urban digital divide; nonlinear effects of institutions.

	(1)	(2)	(3)	(4)	(5)	(6)
Model	*New Member States*		*Controlling for median age*		*Nonlinear impacts of institutions*	
Regional Quality of Governance	−0.0246***	−0.0589***	−0.0258***	−0.0627***	−0.0303***	−0.0450***
	(−3.45)	(−4.09)	(−2.90)	(−4.14)	(−3.27)	(−2.76)
Squared regional Quality of Governance	-	-	-	-	−0.0138***	−0.0264***
					(−3.48)	(−4.42)
Metropolitan regions	0.000380	0.00182	0.000126	0.00259	0.000519	0.00184
	(0.14)	(0.54)	(0.04)	(0.76)	(0.16)	(0.56)
Mountain regions	−0.00351**	0.000760	−0.00361	0.00127	−0.00355	−0.00208
	(−2.54)	(0.22)	(−1.59)	(0.35)	(−1.60)	(−0.70)
Island regions	0.00130	−0.0110	0.00217	−0.0122	0.00485	0.00327
	(0.56)	(−1.41)	(0.46)	(−1.50)	(1.09)	(0.43)
Sparsely populated regions	0.0473***	0.0168	0.0486***	0.0110	0.0422**	0.0245
	(3.32)	(0.72)	(2.70)	(0.45)	(2.19)	(1.12)
Border regions	−0.00109	0.00213*	−0.000238	0.00245*	0.00148	0.00384
	(−0.58)	(1.77)	(−0.14)	(1.84)	(0.88)	(1.28)
R&D intensity	−0.267	0.187	−0.0233	0.189	0.305	0.657
	(−1.03)	(0.51)	(−0.06)	(0.51)	(0.79)	(1.53)
Capital regions	−0.0119	−0.0177	−0.00898	−0.0197*	−0.00573	−0.00632
	(−1.33)	(−1.64)	(−0.82)	(−1.76)	(−0.57)	(−0.64)
Population density	0.00125	−0.0207**	−0.00846	−0.0228**	−0.00137	00000695
	(0.13)	(−2.20)	(−0.95)	(−2.22)	(−0.14)	(0.07)
Share of artificial land	−0.0000581	−0.000329	−0.000105	−0.000287	−0.000194	−0.000299
	(−0.26)	(−1.04)	(−0.35)	(−0.90)	(−0.67)	(−0.98)
Tertiary education	0.00107*	0.00216***	0.00129*	0.00236***	0.00139**	0.00184***
	(1.83)	(2.74)	(1.83)	(3.08)	(2.30)	(2.87)
CEECs	−0.0108	−0.0471*	−0.00118	−0.0505*	−0.00375	−0.0196
	(−0.83)	(−1.70)	(−0.06)	(−1.76)	(−0.20)	(−0.78)
Median population age	-	-	−0.174	0.985	−0.474	
			(−0.14)	(0.75)	(−0.41)	
Constant term	0.248***	0.227***	0.251***	0.178***	0.263***	0.233***
	(19.63)	(13.98)	(4.23)	(2.66)	(4.79)	(14.43)
Country FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	180	127	127	127	127	127
Estimation method	OLS	IV	OLS	IV	OLS	IV
Instruments used	-	Trajan-period network of Roman roads; 1870 European railways; regional trust	-	Trajan-period network of Roman roads; 1870 European railways; regional trust	-	Trajan-period network of Roman roads; 1870 European railways; regional trust (linear and squared)
Adjusted R^2	0.272	0.220	0.317	0.191	0.394	0.319

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

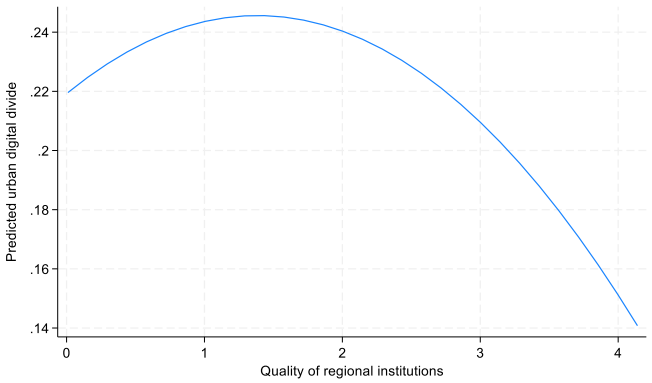


Fig. 4. Nonlinear effects of the quality of regional institutions on the urban digital divide. Source: Authors' elaboration.

not be strong enough to instrument both the linear and the squared term. Moreover, Fig. 4 shows graphically the marginal effects of regional institutions on the digital divide as the quality of institutions increases.[21]

6. Conclusions

This paper addressed a relevant, and, to date, little inspected issue in the diffusion of digital technologies, viz. its institutional determinants at the local level. To this aim, we assembled data from micro and aggregate sources for a sample of 181 European cities, covering the whole EU27 plus the UK. To the best of our knowledge, this is the first paper focusing explicitly on the role of regional institutional quality in driving the within-city digital divide, with a specific discussion of the causal mechanisms behind this relationship. Historical instruments and data on soft institutions allow to safely infer that regional institutional quality causes a decrease in the extent to which digital technologies are unequally distributed and used among urban dwellers.

These results may bear important normative implications. The space for local policymakers to modify local incentives for the diffusion of digital technologies is in fact rather large. At the same time, acting on local institutions is a rather complex undertaking. As the evidence suggests, two issues appear on this front problematic. On the one hand, regional institutional quality depends heavily on national institutional contexts (Rodríguez-Pose, 2013). On the other hand, the investment in improving the quality of institutions usually takes a significant amount of time to pay off (North, 1990). Within this framework, additional elements of complexity stem from the standard problem of the imperfect overlapping of the time needed to ignite long-run change, often at odds with the short-run nature of political cycles (analytically modelled in Acemoglu et al. (2021).

From a practitioner's perspective, understanding the causal impact of the quality of institutions on the extent of the digital divide amongst citizens can be used by decision makers in situations where the lack of empirical evidence on the digital gaps makes gauging the gravity of the phenomenon difficult. In an area with lower institutional quality, for which information is more readily available than for the actual level of the digital divide, policy makers might expect a significant problem related to digital skills and may plan corrective measures accordingly.

While credible financial estimates of the revenues missed due to the digital divide are not available, indirect evidence does suggest the relevance of addressing this issue. Two main arguments appear substantive. Firstly, the pervasive diffusion of digital technologies, especially in tertiary activities, already in place before 2020, but vastly accelerated by the COVID-19 pandemic (Stokel-Walker, 2022). Secondly, the substantial investment in bridging the digital divide that experts suggest policymakers to make (see e.g. the policy recommendations to the Biden-Harris administration in Chakravorti (2021), calling for 240 Billion USD to be spent only on bridging the broadband infrastructure digital divide in the US).

As institutional change is costly and associated with long adjustment times (Raya and Resosudarmo, 2024), a cost-benefit analysis of an improvement of local institutions to also reduce the urban digital divide would be a natural future research avenue for our work.

CRediT authorship contribution statement

Andrea Caragliu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chiara F. Del Bo:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

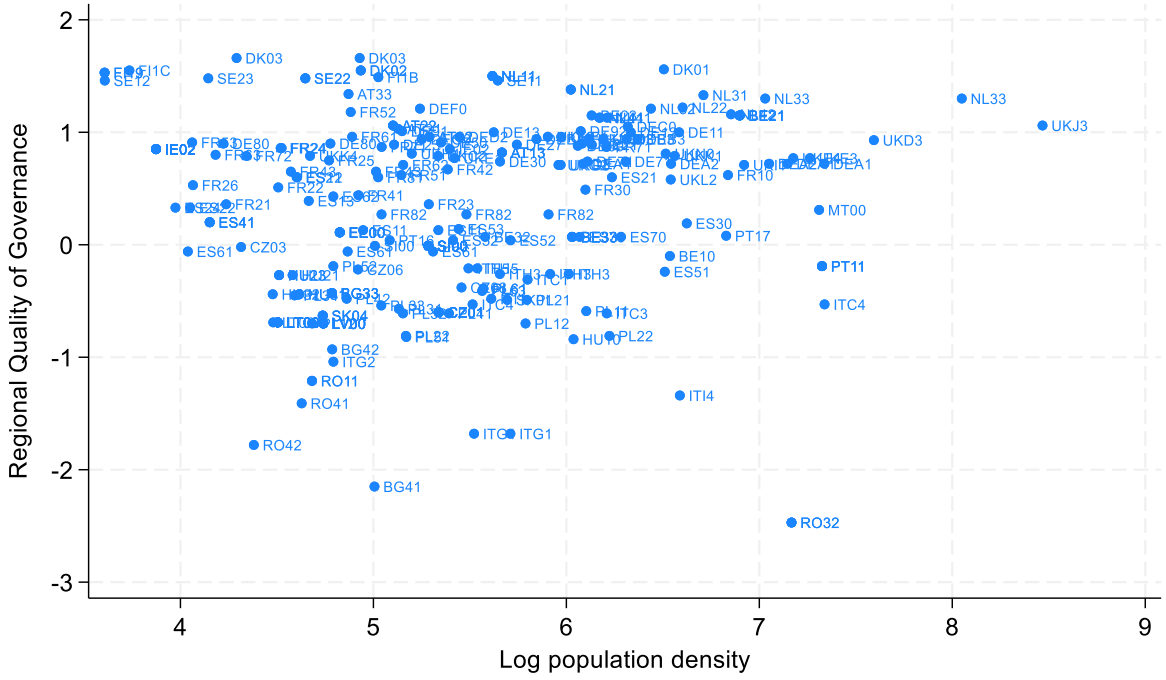
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Technical Appendix A.1. Within-NUTS2 variation in regional quality of institutions between NUTS3 regions

The argument concerning the limited spatial variation characterizing regional quality of institutions between NUTS3 regions belonging to the same NUTS2 area can be indirectly be supported by looking at how NUTS-level scores in institutions vary with population density. If the former increases with the latter, one would logically expect core NUTS3 regions to be characterized by better institutions that peri-urban areas. Figure A.1 seems to suggest that this is not the case.

[21] Throughout IV estimates, all standard IV tests suggest the rejection of the weak instrument test, of the weak identification test, while we do not reject the null hypothesis of the overid test (with a few exceptions for nonlinear effects of the quality of regional institutions). See Technical Appendix A.3. IV tests for further details.



FigureA.1. Scatter plot of regional quality of governance (Y-axis) and log population density (X-axis)

This argument can be further strengthened by regressing the quality of regional institutions on log population density (Column 1) and settlement structure dummy variables (Column 2), as shown in Table A.1.

Once again, the evidence suggests that there is no statistically significant association between the urban structure of a region and its quality of institutions.

Table A.1
Main descriptive statistics of the urban digital divide and quality of regional institutions (overall and by Country)

Model	*(1)*	*(2)*
Dep. variable	*Quality of regional institutions*	
Log population density	0.0912	
	(1.47)	
Agglomerated regions		0.0393
		(0.16)
Urban regions		−0.0348
		(−0.15)
Constant term	−0.222	0.282
	(−0.66)	(1.23)
Observations	251	254
Adjusted R^2	0.007	−0.006

t statistics in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Technical Appendix A.2. List of urban areas in the data set

Table A.2
List of urban areas in the data set

City name	Country
Wien	AT
Graz	AT
Linz	AT
Salzburg	AT
Innsbruck	AT
Bruxelles / Brussel	BE
Antwerpen	BE
Gent	BE
Charleroi	BE
Liége	BE
Sofia	BG

(*continued on next page*)

Table A.2 (*continued*)

City name	Country
Plovdiv	BG
Varna	BG
Praha	CZ
Brno	CZ
Ostrava	CZ
Plzen	CZ
Berlin	DE
Hamburg	DE
München	DE
Köln	DE
Frankfurt am Main	DE
Stuttgart	DE
Dresden	DE
Düsseldorf	DE
Bremen	DE
Hannover	DE
Nürnberg	DE
Bielefeld	DE
Wiesbaden	DE
Göttingen	DE
Darmstadt	DE
Freiburg im Breisgau	DE
Regensburg	DE
Schwerin	DE
Erfurt	DE
Augsburg	DE
Bonn	DE
Karlsruhe	DE
Mönchengladbach	DE
Mainz	DE
Kiel	DE
Saarbrucken	DE
Koblenz	DE
Rostock	DE
Köbenhavn	DK
Aarhus	DK
Odense	DK
Aalborg	DK
Tallinn	EE
Madrid	ES
Barcelona	ES
Valencia	ES
Sevilla	ES
Zaragoza	ES
Malaga	ES
Murcia	ES
Las Palmas	ES
Valladolid	ES
Palma de Mallorca	ES
Vitoria/Gasteiz	ES
Oviedo	ES
Santander	ES
Bilbao	ES
Cordoba	ES
Alicante/Alacant	ES
Vigo	ES
Sta. Cruz de Tenerife	ES
Helsinki	FI
Tampere	FI
Turku	FI
Paris	FR
Lyon	FR
Toulouse	FR
Strasbourg	FR
Bordeaux	FR
Nantes	FR
Lille	FR
Montpellier	FR
Saint-Etienne	FR
Rennes	FR
Amiens	FR
Rouen	FR
Nancy	FR
Reims	FR
Orléans	FR
Dijon	FR

(*continued on next page*)

Table A.2 (*continued*)

City name	Country
Poitiers	FR
Clermont-Ferrand	FR
Caen	FR
Limoges	FR
Besançon	FR
Grenoble	FR
Budapest	HU
Miskolc	HU
Pecs	HU
Debrecen	HU
Székesfehérvár	HU
Dublin	IE
Cork	IE
Roma	IT
Milano	IT
Torino	IT
Palermo	IT
Genova	IT
Firenze	IT
Bologna	IT
Catania	IT
Venezia	IT
Verona	IT
Cagliari	IT
Padova	IT
Brescia	IT
Modena	IT
Vilnius	LT
Kaunas	LT
Riga	LV
Valletta	MT
's Gravenhage	NL
Amsterdam	NL
Rotterdam	NL
Utrecht	NL
Eindhoven	NL
Tilburg	NL
Groningen	NL
Enschede	NL
Arnhem	NL
Heerlen	NL
Breda	NL
Warszawa	PL
Lodz	PL
Krakow	PL
Wroclaw	PL
Poznan	PL
Gdansk	PL
Szczecin	PL
Bydgoszcz	PL
Lublin	PL
Katowice	PL
Bialystok	PL
Kielce	PL
Rzeszow	PL
Opole	PL
Czestochowa	PL
Radom	PL
Lisboa	PT
Oporto	PT
Coimbra	PT
Bucuresti	RO
Cluj-Napoca	RO
Timisoara	RO
Craiova	RO
Stockholm	SE
Göteborg	SE
Malmöe	SE
Uppsala	SE
Ljubljana	SI
Maribor	SI
Bratislava	SK
Kosice	SK
London	UK
Birmingham	UK
Leeds	UK

(*continued on next page*)

Table A.2 (*continued*)

City name	Country
Glasgow	UK
Bradford	UK
Liverpool	UK
Edinburgh	UK
Manchester	UK
Cardiff	UK
Sheffield	UK
Bristol	UK
Belfast	UK
Newcastle upon Tyne	UK
Leicester	UK
Cambridge	UK
Exeter	UK
Portsmouth	UK
Coventry	UK
Kingston-upon-Hull	UK
Stoke-on-Trent	UK
Nottingham	UK

Technical Appendix A.3. Variation in the urban digital divide and the quality of regional institutions

Table A.3 shows the main descriptive statistics for the core variables in our dataset – the urban digital divide and the quality of regional institutions. These include the number of observations, the mean, the standard error, the coefficient of variation, and the interquartile range (i.e., the ratio of third to first quartile). We first show a breakdown of these statistics for all EU Countries with the inclusion of the UK, and then at the bottom of Table A.1 show results for the overall database.

Table A.3
Main descriptive statistics of the urban digital divide and quality of regional institutions (overall and by Country)

Country code	Statistics	Urban digital divide	Quality of regional institutions
AT	Number of obs.	9.00	9.00
	Mean	0.22	0.99
	standard error	0.01	0.06
	Coefficient of variation	0.07	0.17
	Interquartile range	0.03	0.24
BE	Number of obs.	11.00	11.00
	Mean	0.20	0.55
	standard error	0.01	0.18
	Coefficient of variation	0.19	1.06
	Interquartile range	0.06	1.08
BG	Number of obs.	6.00	6.00
	Mean	0.21	−0.80
	standard error	0.00	0.28
	Coefficient of variation	0.04	−0.86
	Interquartile range	0.01	0.50
CY	Number of obs.	1.00	0.00
	Mean	0.33	.
	standard error	.	.
	Coefficient of variation	.	.
	Interquartile range	0.00	.
CZ	Number of obs.	8.00	8.00
	Mean	0.19	−0.45
	standard error	0.01	0.08
	Coefficient of variation	0.10	−0.50
	Interquartile range	0.01	0.30
DE	Number of obs.	30.00	28.00
	Mean	0.22	0.89
	standard error	0.00	0.02
	Coefficient of variation	0.08	0.14
	Interquartile range	0.03	0.24
DK	Number of obs.	5.00	5.00
	Mean	0.19	1.60
	standard error	0.01	0.03
	Coefficient of variation	0.07	0.04
	Interquartile range	0.01	0.11
EE	Number of obs.	5.00	5.00
	Mean	0.26	0.11
	standard error	0.01	0.00
	Coefficient of variation	0.10	0.00
	Interquartile range	0.03	0.00
EL	Number of obs.	10.00	0.00
	Mean	0.23	.

(*continued on next page*)

Table A.3 (*continued*)

Country code	Statistics	Urban digital divide	Quality of regional institutions
	standard error	0.01	.
	Coefficient of variation	0.17	.
	Interquartile range	0.04	.
ES	Number of obs.	22.00	22.00
	Mean	0.26	0.19
	standard error	0.01	0.05
	Coefficient of variation	0.11	1.18
	Interquartile range	0.03	0.29
FI	Number of obs.	4.00	4.00
	Mean	0.21	1.53
	standard error	0.01	0.01
	Coefficient of variation	0.05	0.02
	Interquartile range	0.01	0.03
FR	Number of obs.	26.00	26.00
	Mean	0.20	0.65
	standard error	0.00	0.05
	Coefficient of variation	0.12	0.36
	Interquartile range	0.03	0.37
HR	Number of obs.	17.00	0.00
	Mean	0.21	.
	standard error	0.01	.
	Coefficient of variation	0.17	.
	Interquartile range	0.03	.
HU	Number of obs.	7.00	7.00
	Mean	0.23	−0.42
	standard error	0.00	0.08
	Coefficient of variation	0.04	−0.48
	Interquartile range	0.01	0.17
IE	Number of obs.	8.00	8.00
	Mean	0.19	0.85
	standard error	0.01	0.00
	Coefficient of variation	0.12	0.00
	Interquartile range	0.04	0.00
IT	Number of obs.	15.00	14.00
	Mean	0.22	−0.67
	standard error	0.00	0.14
	Coefficient of variation	0.05	−0.80
	Interquartile range	0.01	0.78
LT	Number of obs.	10.00	10.00
	Mean	0.18	−0.69
	standard error	0.00	0.00
	Coefficient of variation	0.05	0.00
	Interquartile range	0.01	0.00
LU	Number of obs.	1.00	0.00
	Mean	0.22	.
	standard error	.	.
	Coefficient of variation	.	.
	Interquartile range	0.00	.
LV	Number of obs.	6.00	6.00
	Mean	0.20	−0.70
	standard error	0.00	0.00
	Coefficient of variation	0.04	0.00
	Interquartile range	0.01	0.00
MT	Number of obs.	1.00	1.00
	Mean	0.29	0.31
	standard error	.	.
	Coefficient of variation	.	.
	Interquartile range	0.00	0.00
NL	Number of obs.	14.00	14.00
	Mean	0.10	1.30
	standard error	0.01	0.04
	Coefficient of variation	0.33	0.11
	Interquartile range	0.06	0.22
PL	Number of obs.	17.00	17.00
	Mean	0.19	−0.56
	standard error	0.01	0.04
	Coefficient of variation	0.13	−0.31
	Interquartile range	0.01	0.25
PT	Number of obs.	5.00	5.00
	Mean	0.23	−0.09
	standard error	0.01	0.06
	Coefficient of variation	0.09	−1.53
	Interquartile range	0.01	0.23
RO	Number of obs.	8.00	6.00
	Mean	0.23	−1.76
	standard error	0.01	0.24

(*continued on next page*)

Table A.3 (*continued*)

Country code	Statistics	Urban digital divide	Quality of regional institutions
	Coefficient of variation	0.08	−0.34
	Interquartile range	0.03	1.26
SE	Number of obs.	8.00	5.00
	Mean	0.13	1.47
	standard error	0.01	0.00
	Coefficient of variation	0.12	0.01
	Interquartile range	0.02	0.02
SI	Number of obs.	12.00	12.00
	Mean	0.24	−0.01
	standard error	0.01	0.00
	Coefficient of variation	0.09	0.00
	Interquartile range	0.01	0.00
SK	Number of obs.	4.00	4.00
	Mean	0.22	−0.60
	standard error	0.01	0.04
	Coefficient of variation	0.05	−0.12
	Interquartile range	0.01	0.07
UK	Number of obs.	20.00	21.00
	Mean	0.21	0.82
	standard error	0.00	0.03
	Coefficient of variation	0.08	0.14
	Interquartile range	0.02	0.16
Total	*Number of obs.*	*290.00*	*254.00*
	Mean	*0.21*	*0.28*
	standard error	*0.00*	*0.05*
	Coefficient of variation	*0.20*	*2.91*
	Interquartile range	*0.04*	*1.29*

Table A.3 documents a rather significant role played by within Countries interregional differences for both variables.

In particular, for quality of regional institutions, the overall interquartile range for all data is equal to 1.29, but for some Countries the same statistic is equal to 1.26 (Romania), thereby explaining 98 per cent of total regional variation. By the same token, the overall coefficient of variation for the whole data set is equal to 2.91, while in the Iberic peninsula Spain scores 1.18 (41 per cent of total variation), Belgium (1.06; 36 per cent of total variation), and Portugal (-) 1.53 (53 per cent of total variation). These statistics corroborate the statement that within Countries variations in the quality of regional institutions are expected to play a major role in explaining the urban digital divide.

The relationship between the urban digital divide and regional quality of governance is shown in Figure A.2.

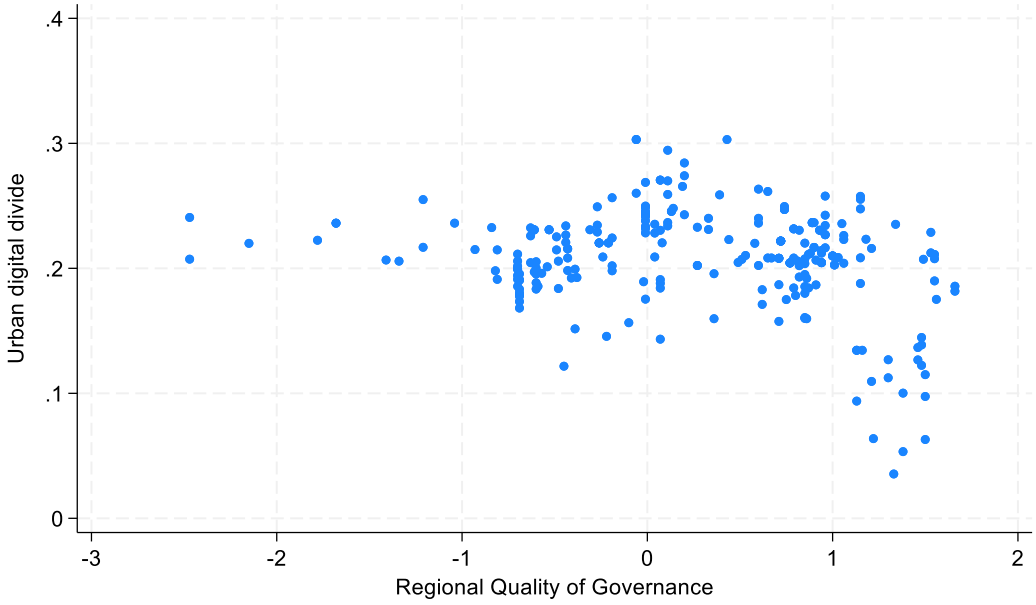


Figure A.2. Scatter plot of urban digital divide and regional quality of governance

Similar conclusions emerge from the analysis of Figures A.2 and A.3, showing the spatial distribution of the urban digital divide, and regional quality of institutions, respectively.

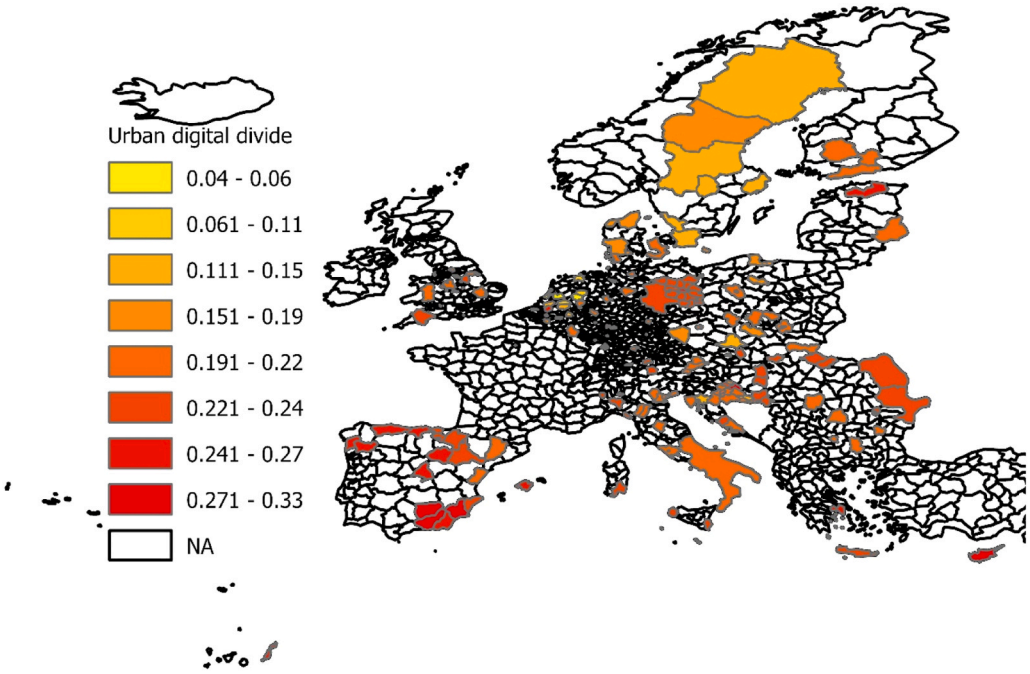


Figure A.2. Spatial distribution of the urban digital divide

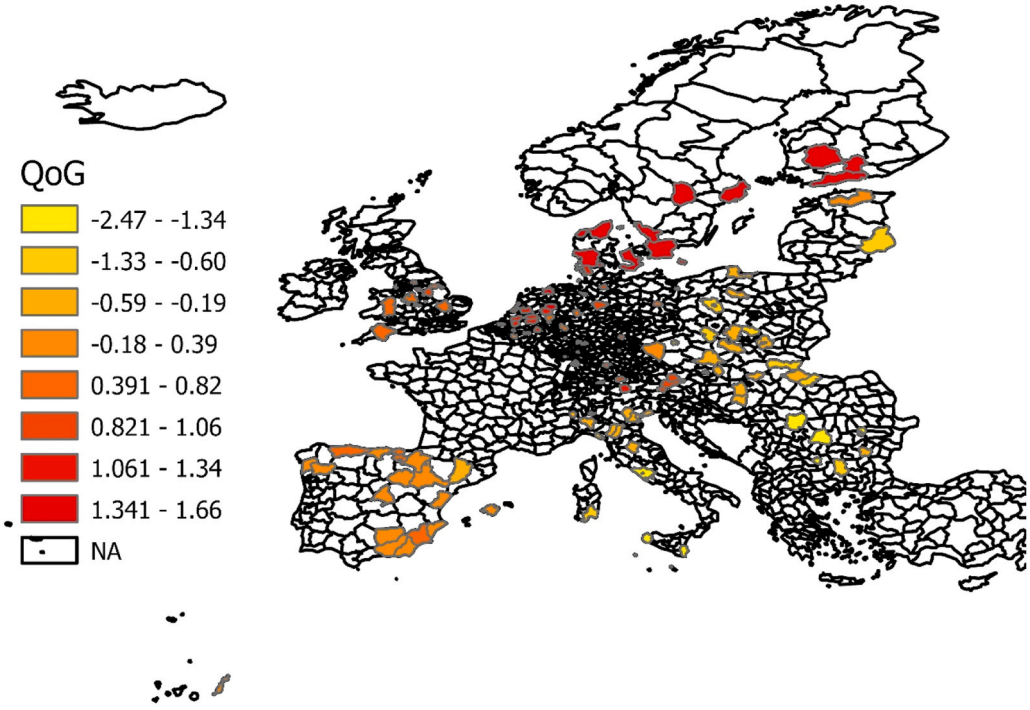


Figure A.3. Spatial distribution of the quality of regional institutions

The interesting pattern emerging from comparing the two maps is that not necessarily regions benefitting from high-quality institutions (e.g., areas in the Scandinavian peninsula) are automatically associated with low levels of urban digital divide. The geography of these two mechanisms is rather complex, and calls in fact for the analyses discussed in this paper.

Technical Appendix A.4. Trust as a function of geodesic coordinates

Table A.4
Trust as a function of geodesic coordinates

Dep. variable	Trust
Latitude	−0.00405***
	(−2.95)
Longitude	0.00855***
	(4.09)
Constant	−0.0471
	(−0.47)
Observations	132
Adjusted R^2	0.119

t statistics in parentheses. * p < 0.1, ** p < 0.05, *** p < 0.01

Technical Appendix A.5. IV tests

A.5.1. Baseline specification

Table A.5
First stage regression results and underidentification, weak identification, and overidentification test results for baseline specifications (Table 3)

	(1)	(2)	(3)	(4)
Model	*Quality of governance*	*Quality*	*Impartiality*	*Corruption*
Distance from 1870 railways	−0.001***	−0.001***	−0.001**	−0.001**
	(0.0003)	(0.0003)	(0.0004)	(0.0003)
Distance from Trajan roman road network	0.001	0.004***	0.003*	0.003*
	(0.001)	(0.001)	(0.001)	(0.001)
Trust	4.01***	4.38***	4.38***	5.33***
	(0.26)	(0.33)	(0.33)	(0.35)
Constant term	0.218***	−1.52 ***	−1.61 ***	0.201***
	(60.78)	(0.14)	(0.14)	(38.59)
Observations	172	107	107	107
Centered R^2	0.50	0.59	0.53	0.60
Uncentered R^2	0.55	0.60	0.55	0.61
Joint F-test pf excluded instruments	84.26***	65.45***	59.76***	100.26***
Underidentification test (Kleibergen-Paap rk LM statistic)	50.81***	33.80***	37.02***	37.77***
	[0.000]	[0.000]	[0.000]	[0.000]
Weak identification test (Cragg-Donald Wald F-statistic)	56.68***	49.85***	38.77***	52.36***
Stock-Yogo weak ID test critical value for 5 % maximal IV relative bias	13.91	13.91	13.91	13.91
Overidentification test (Hansen's J statistic)	1.20	1.83	1.20	1.91
	[0.55]	[0.40]	[0.55]	[0.39]
Endogeneity test	21.12***	8.477***	15.42***	9.90***
	[0.000]	[0.000]	[0.000]	[0.000]

t statistics in parentheses, p-values in square brackets. * p < 0.1, ** p < 0.05, *** p < 0.01

A.5.2. Robustness checks

Table A.6
First stage regression results and underidentification, weak identification, and overidentification test results for robustness checks (Table 5)

	(1)	*(2)*	*(3)*	*(4)*	*(5)*
Model	*Controlling for regional typologies*	*Controlling for location characteristics*	*Controlling for human capital*	*Controlling for urban GDP*	*Controlling for Country heterogeneity*
Distance from 1870 railways	−0.001*	−0.002***	−0.002***	0.003	−0.001
	(0.0005)	(0.0001)	(0.0005)	(0.006)	(0.001)
Distance from Trajan roman road network	0.002	0.002	0.001	0.003*	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Trust	4.07***	2.80***	2.30***	1.36***	1.52***
	(0.33)	(0.33)	(0.34)	(0.37)	(0.40)
Constant term	−1.25***	−0.95***	−1.43***	−8.04***	−7.22***
	(0.26)	(0.23)	(0.26)	(1.13)	(1.24)
Observations	131	127	127	127	127
Centered R^2	0.59	0.73	0.77	0.84	0.85
Uncentered R^2	0.63	0.76	0.80	0.86	0.87
Joint F-test pf excluded instruments	37.26***	26.77***	40.34***	81.28***	89.41***
Underidentification test (Kleibergen-Paap rk LM statistic)	40.87***	24.17***	28.31***	17.24***	17.57***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Weak identification test (Cragg-Donald Wald F-statistic)	47.34***	32.41***	26.53***	9.11**	10.21**

(*continued on next page*)

Table A.6 (*continued*)

Model	*(1)* *Controlling for regional typologies*	*(2)* *Controlling for location characteristics*	*(3)* *Controlling for human capital*	*(4)* *Controlling for urban GDP*	*(5)* *Controlling for Country heterogeneity*
Stock-Yogo weak ID test critical value for 5 % maximal IV relative bias	13.91	13.91	13.91	13.91	13.91
Overidentification test (Hansen's J statistic)	0.21	4.36	6.98**	4.21	4.47
	[0.90]	[0.11]	[0.03]	[0.12]	[0.11]
Endogeneity test	21.81***	14.63***	10.25***	17.30***	11.92***
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]

t statistics in parentheses, p-values in square brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A.5.3. Nonlinear effects

Table A.6
First stage regression results and underidentification, weak identification, and overidentification test results for nonlinear effects (Table 6)

Model	*(1)* *New Member States*	*(2)* *Median age*	*(3)* *Nonlinear impacts of institutions*
Distance from 1870 railways	−0.001**	−0.001**	0.008*
	(0.0005)	(0.0005)	(0.004)
Distance from Trajan roman road network	0.004**	0.003**	0.001
	(0.001)	(0.001)	(0.001)
Trust	1.60***	1.60***	−7.07*
	(0.45)	(0.44)	(3.66)
Squared distance from 1870 railways	-	-	−0.001
			(0.000)
Squared distance from Trajan roman road network	-	-	−0.002*
			(0.001)
Squared trust	-	-	12.75***
			(4.07)
Constant term	−0.68**	−1.67**	1.07
	(0.27)	(0.76)	(0.84)
Observations	127	127	127
Centered R^2	0.83	0.83	0.53
Uncentered R^2	0.85	0.85	0.73
Joint F-test pf excluded instruments	64.99***	64.30***	17.60***
Underidentification test (Kleibergen-Paap rk LM statistic)	22.06***	40.87***	11.61**
	[0.000]	[0.000]	[0.04]
Weak identification test (Cragg-Donald Wald F-statistic)	14.01***	13.00**	6.99*
Stock-Yogo weak ID test critical value for 5 % maximal IV relative bias	13.91	13.91	13.91
Overidentification test (Hansen's J statistic)	9.92***	9.41***	20.49***
	[0.00]	[0.00]	[0.00]
Endogeneity test	5.04**	5.54**	5.08**
	[0.01]	[0.01]	[0.01]

t statistics in parentheses, p-values in square brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

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